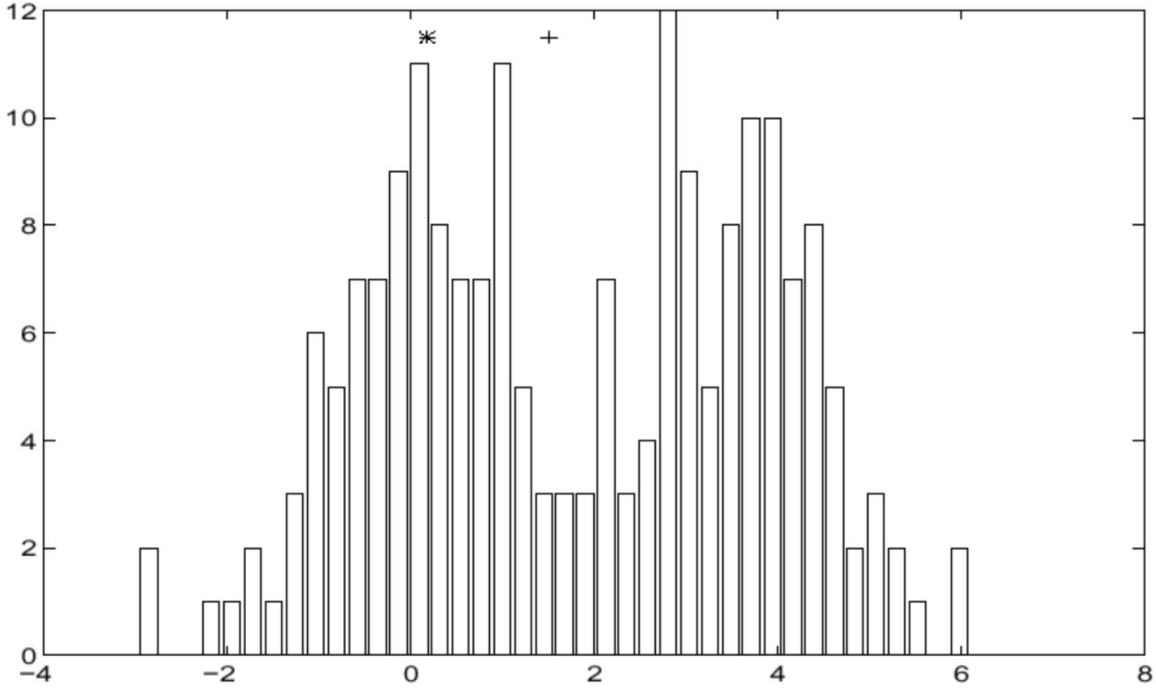
Mean-Shift

Федоров Вячеслав, группа 25222

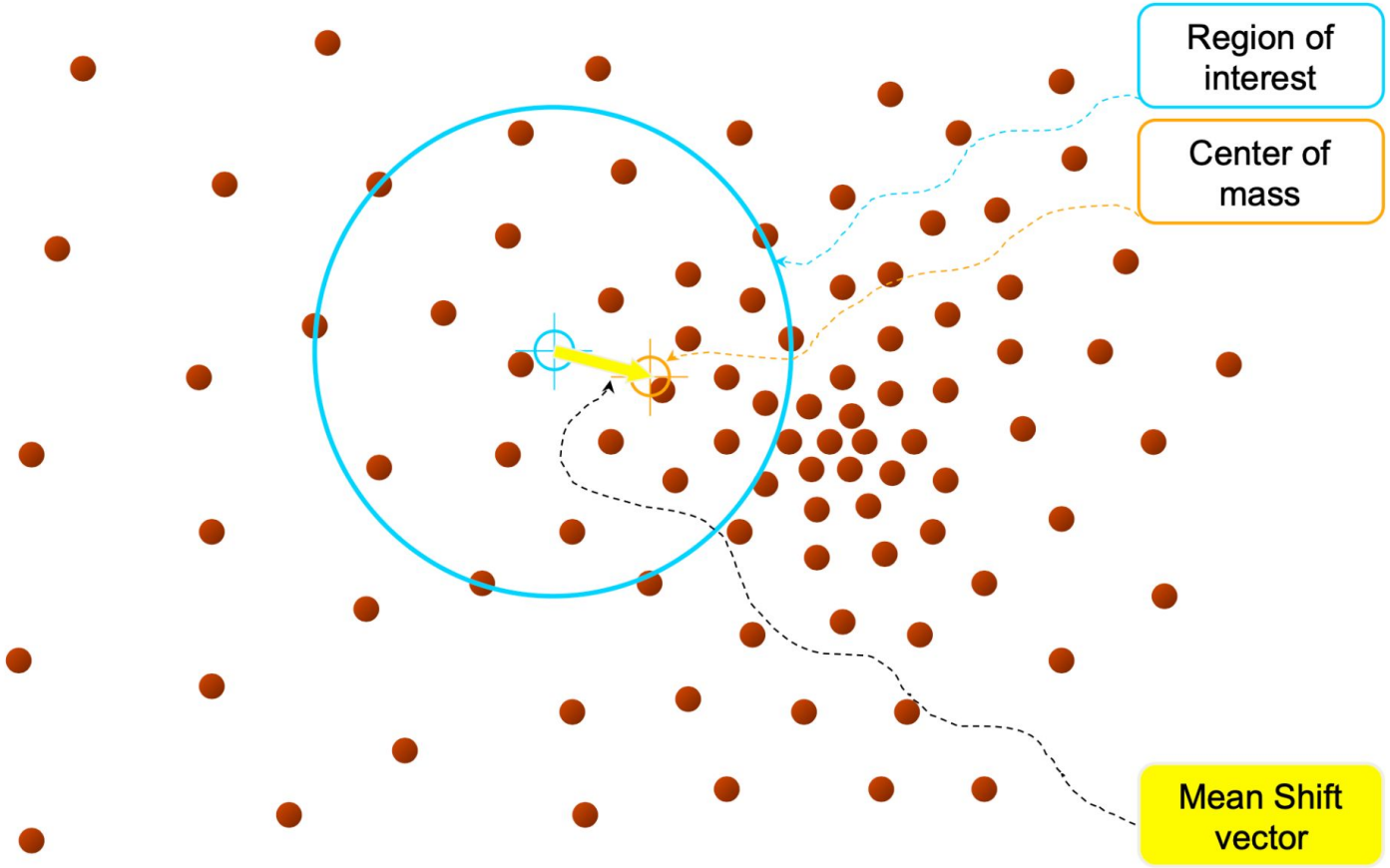
По материалам статьи Comaniciu и Meer 2002 года "Mean Shift: A Robust Approach toward Feature Space Analysis"

Mean-Shift Algorithm



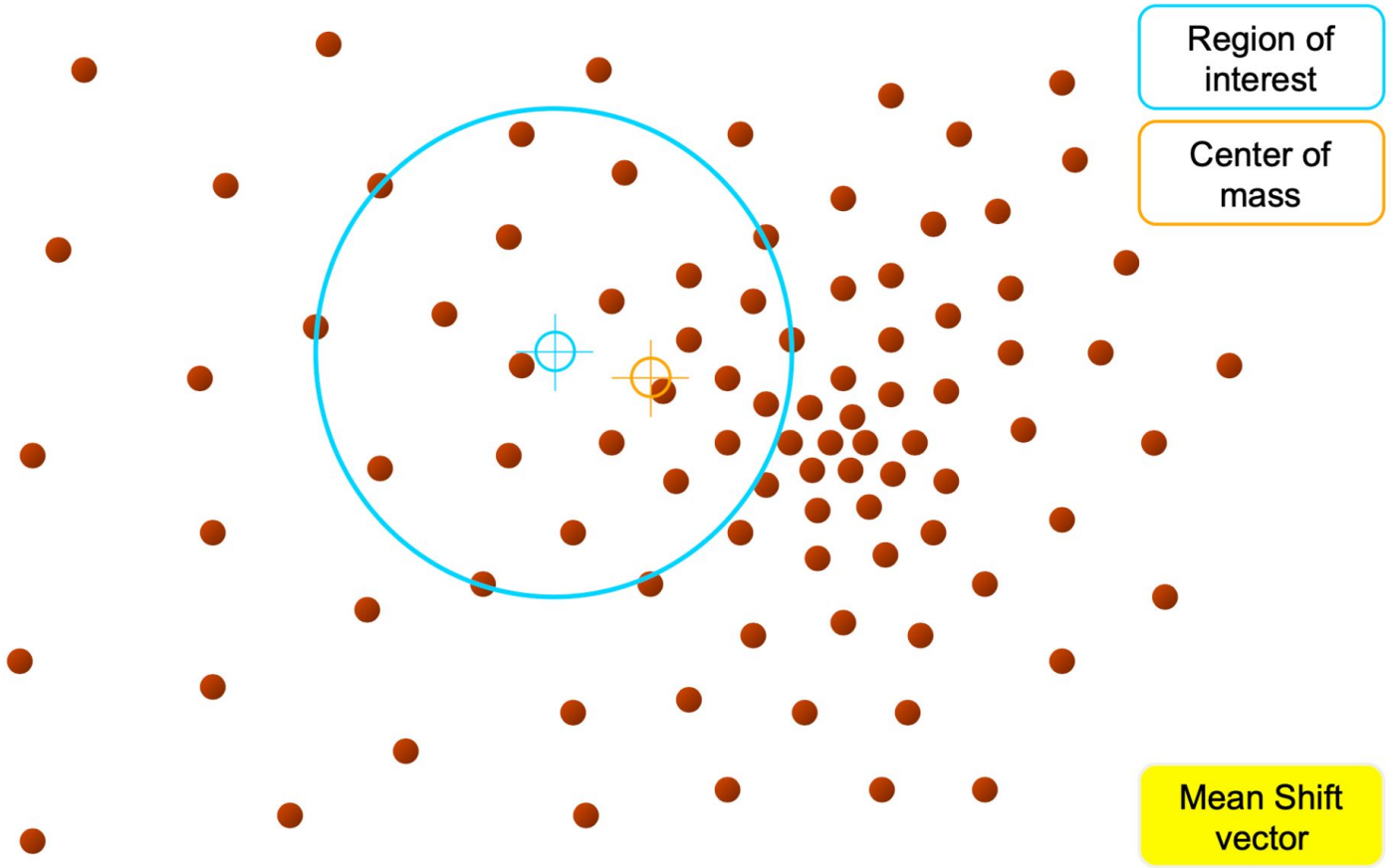
- Iterative Mode Search
  1. Initialize random seed, and window W
  2. Calculate center of gravity (the "mean") of W: $\sum_{x \in W} x H(x)$
  3. Shift the search window to the mean
  4. Repeat Step 2 until convergence

Mean-Shift



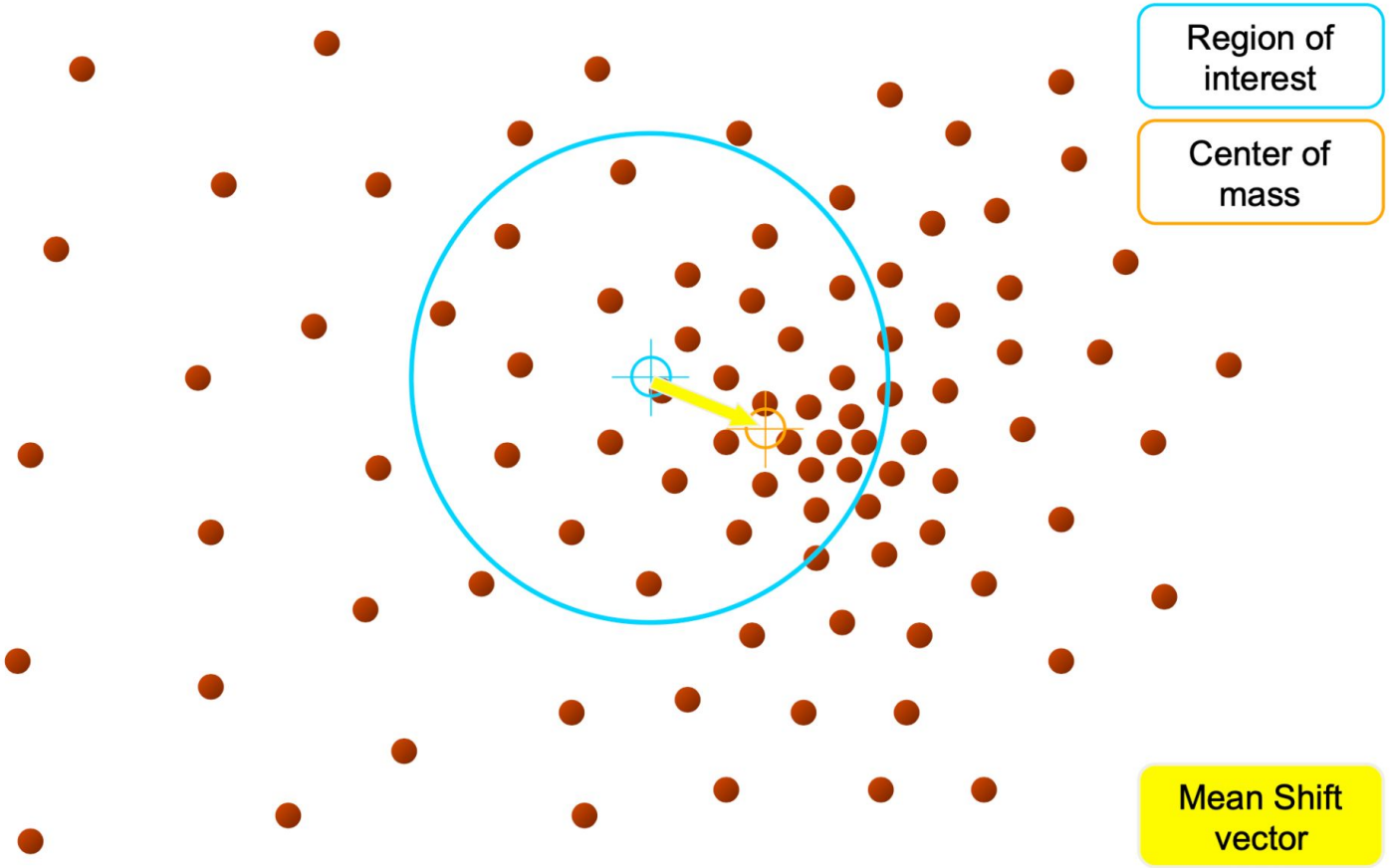
Slide by Y. Ukrainitz & B. Sarel

Mean-Shift



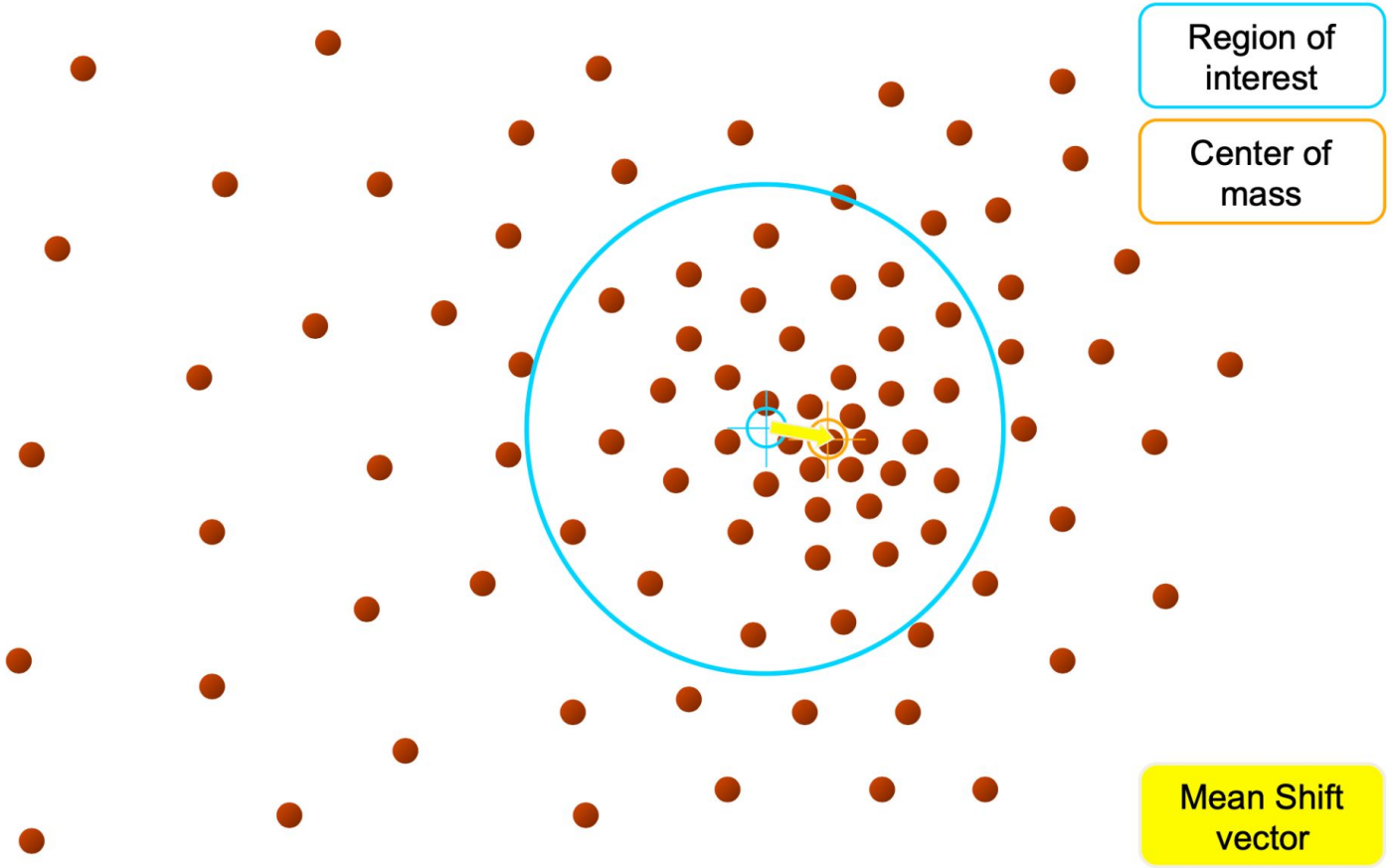
Slide by Y. Ukrainitz & B. Sarel

Mean-Shift



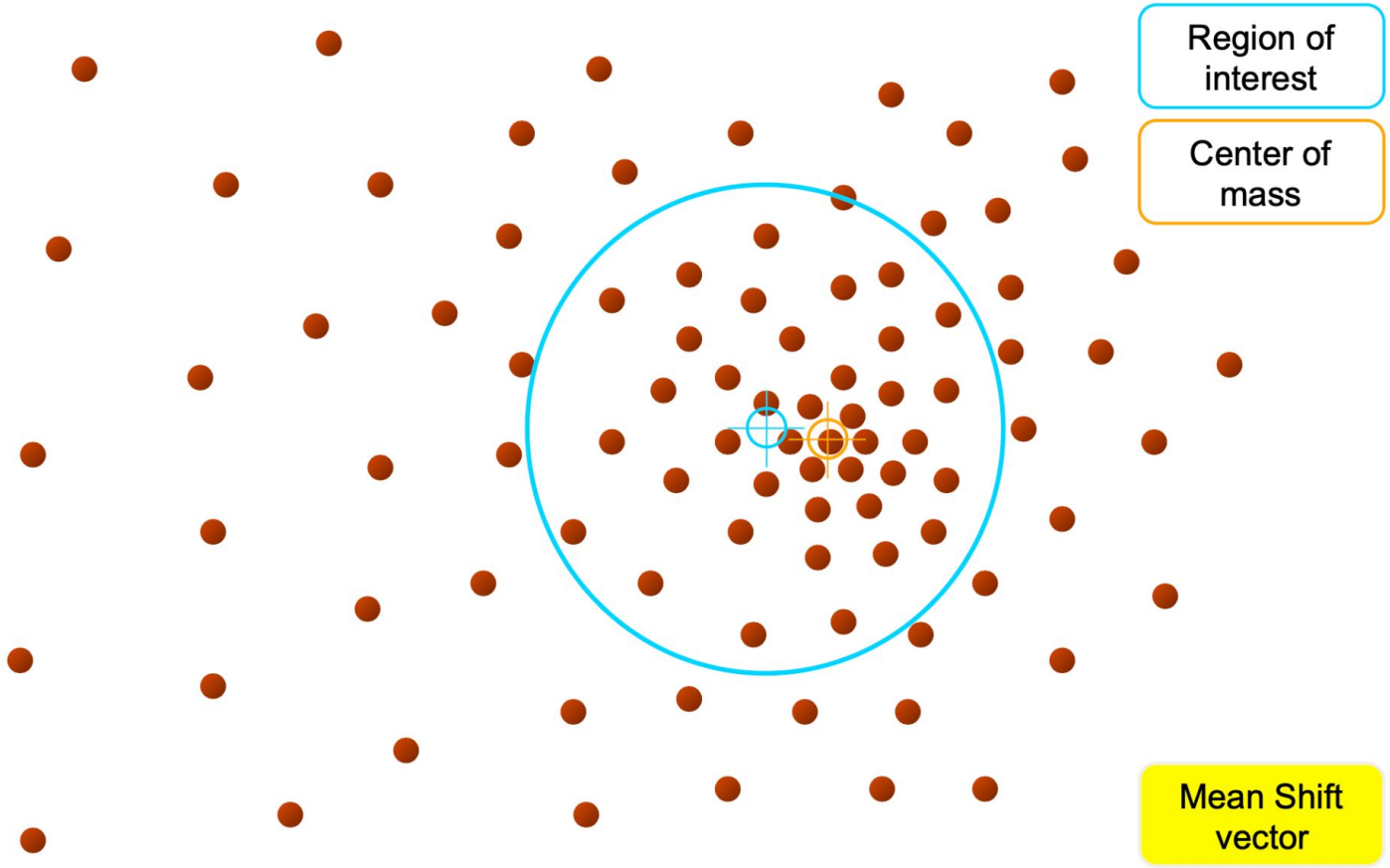
Slide by Y. Ukrainitz & B. Sarel

Mean-Shift



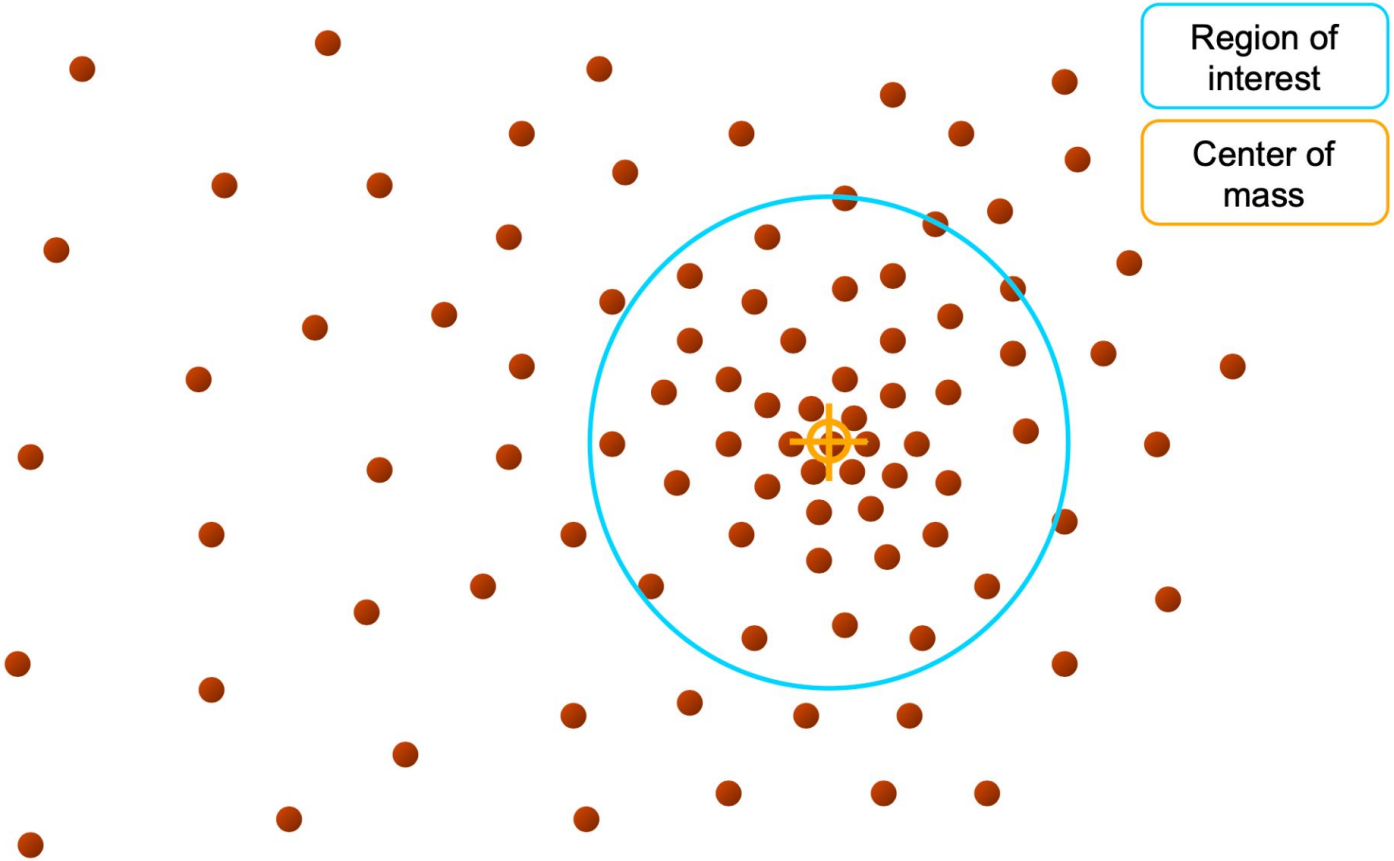
Slide by Y. Ukrainitz & B. Sarel

Mean-Shift



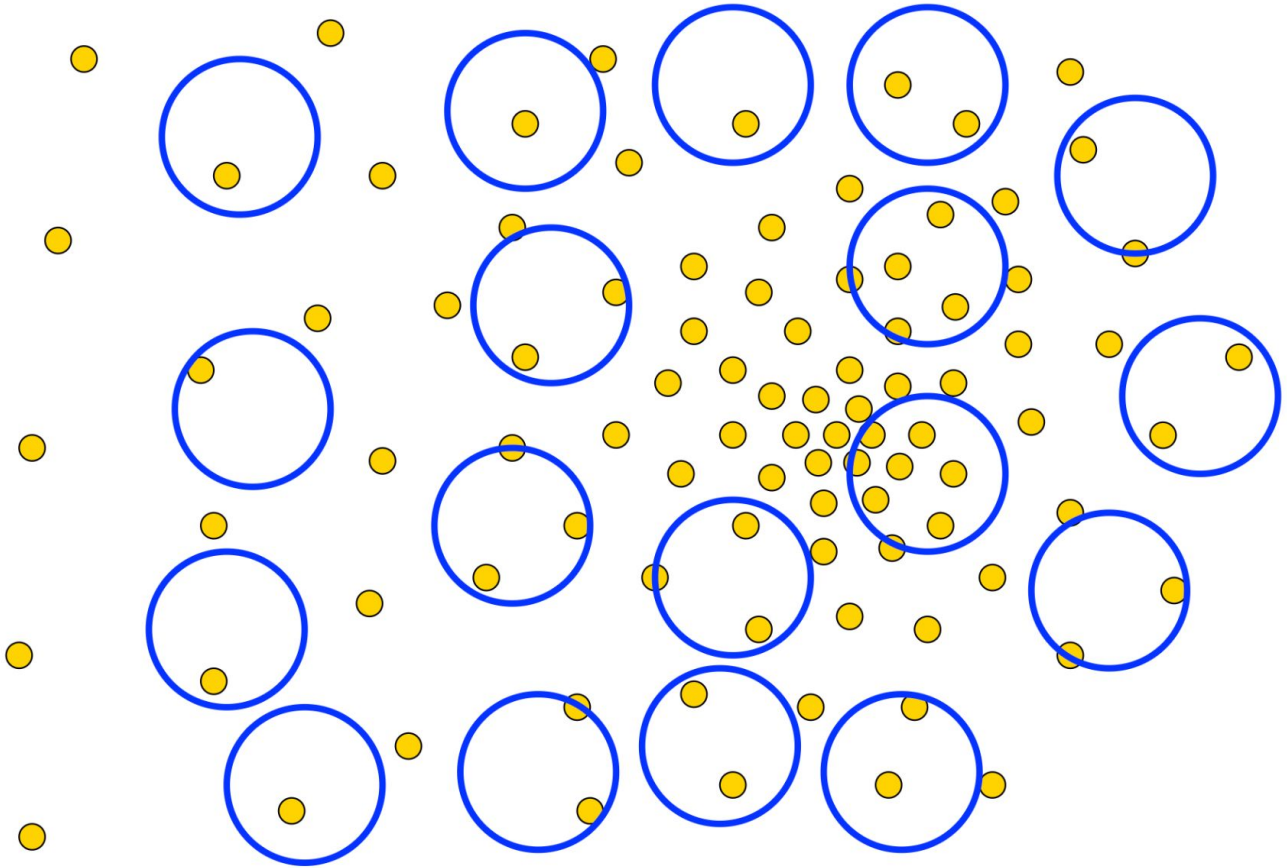
Slide by Y. Ukrainitz & B. Sarel

Mean-Shift



Slide by Y. Ukrainitz & B. Sarel

Real Modality Analysis

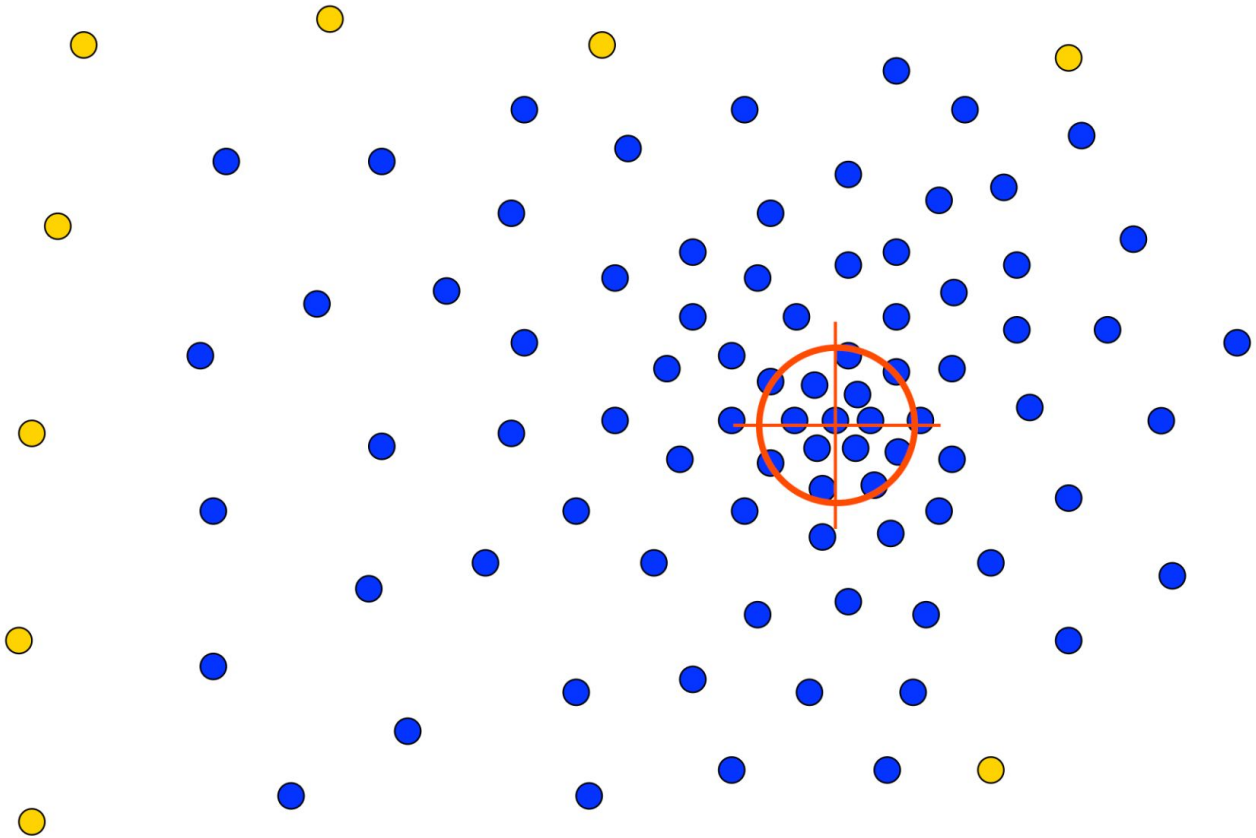


Tessellate the space with windows

Run the procedure in parallel

Slide by Y. Ukrainitz & B. Sarel

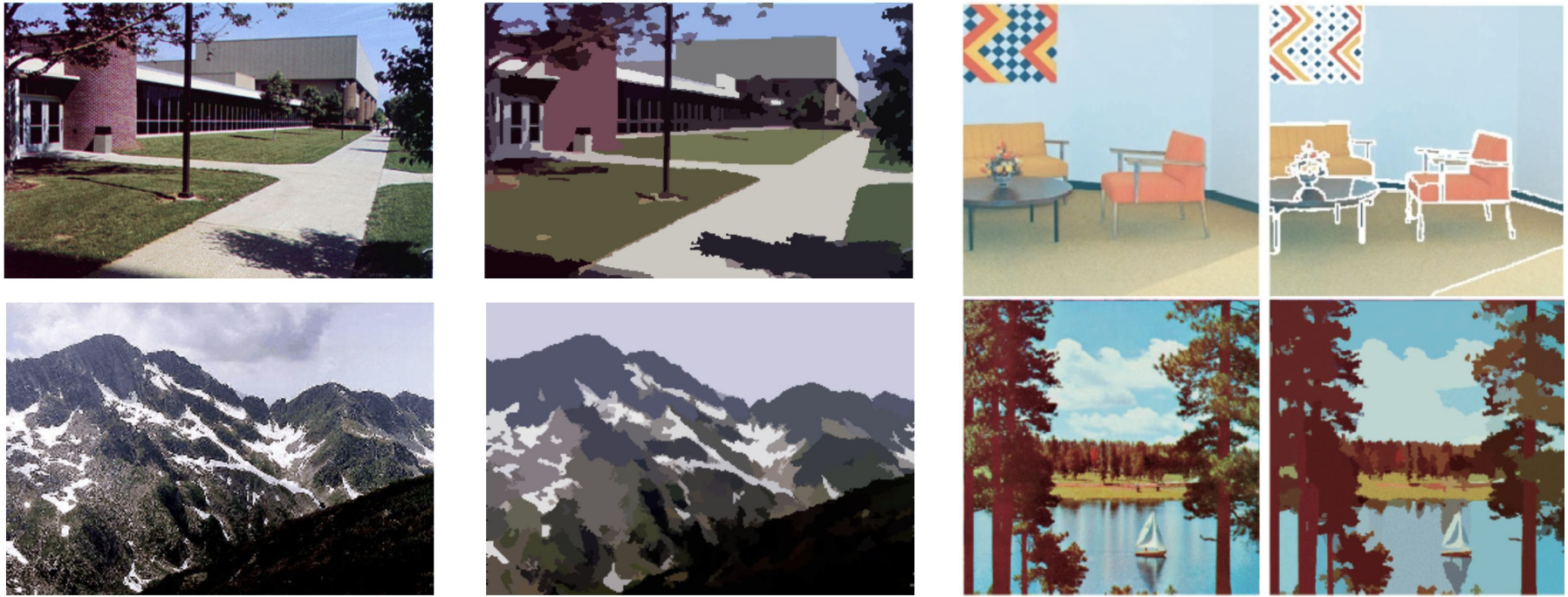
Real Modality Analysis



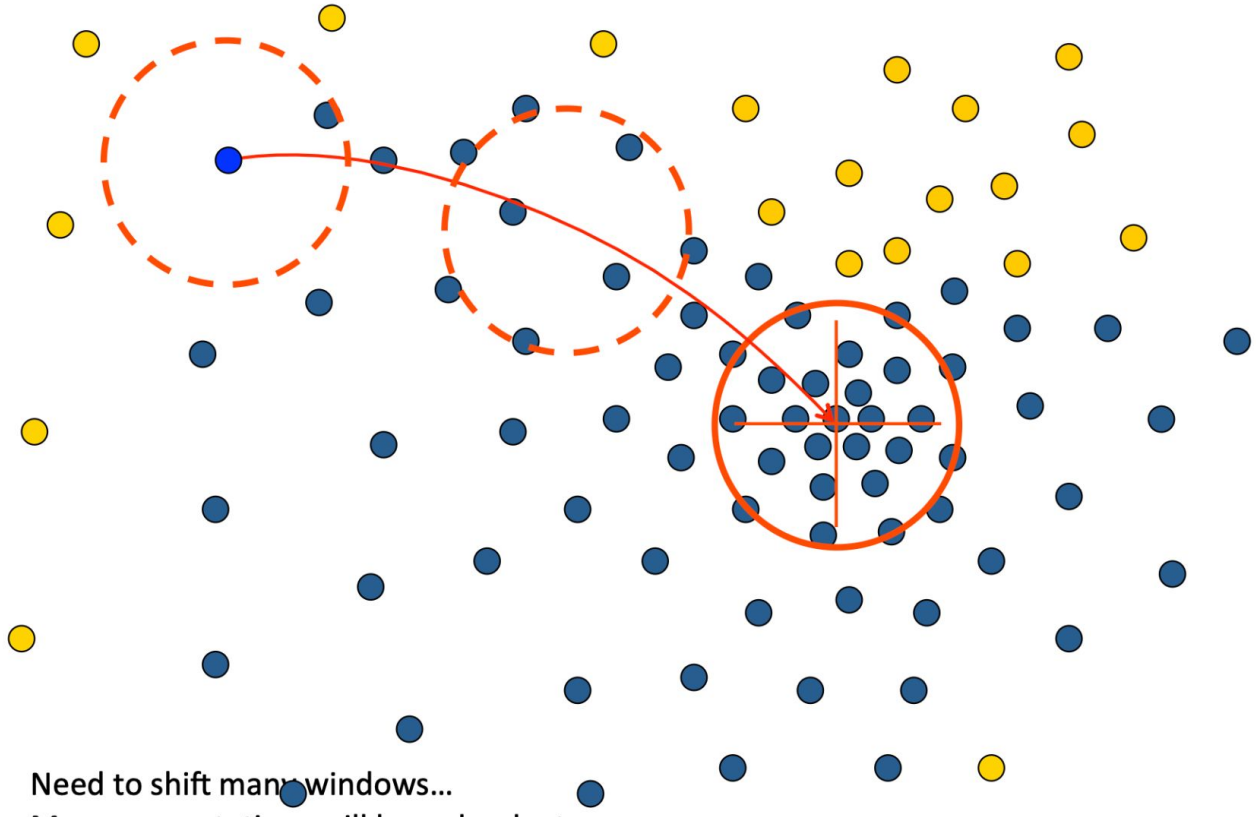
The blue data points were traversed by the windows towards the mode.

Slide by Y. Ukrainitz & B. Sarel

Mean-Shift Clustering/Segmentation



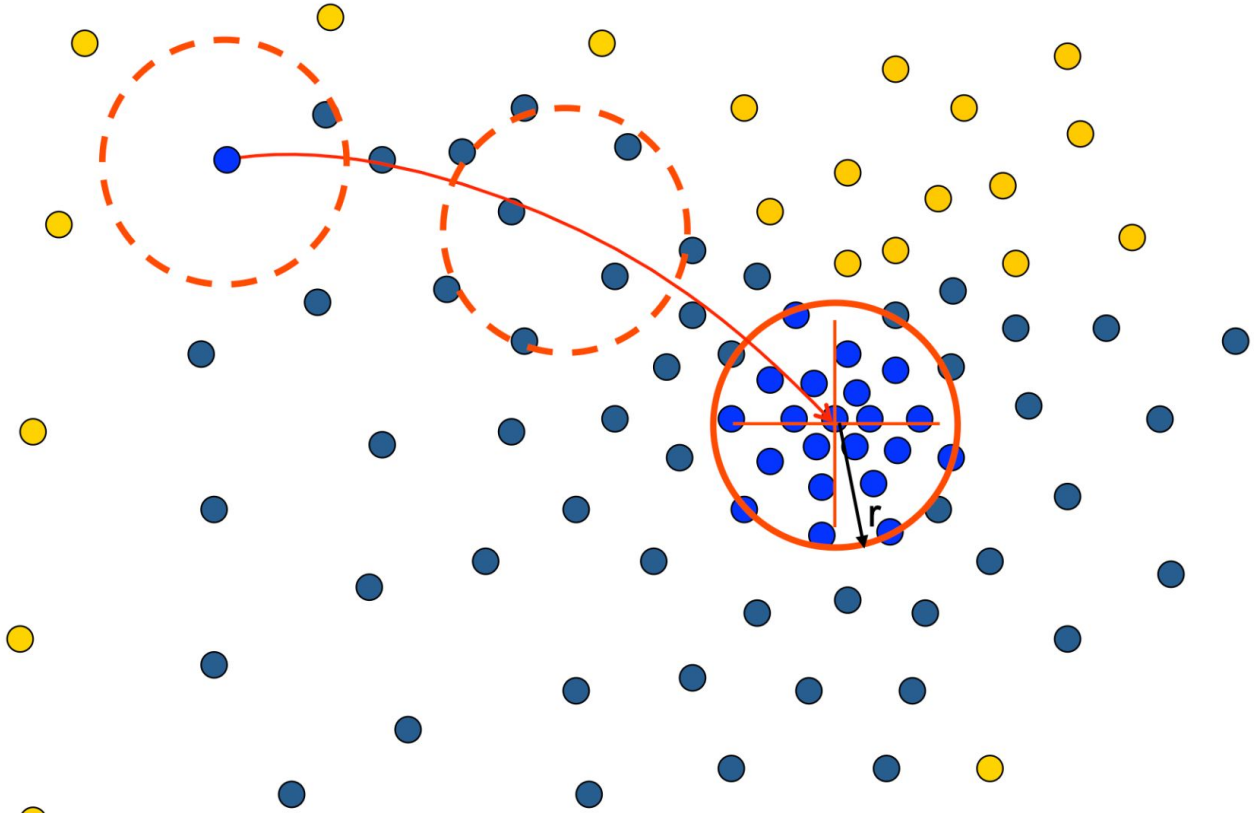
Problem: Computational Complexity



- Need to shift many windows…
- Many computations will be redundant.

Slide credit: Bastian Leibe

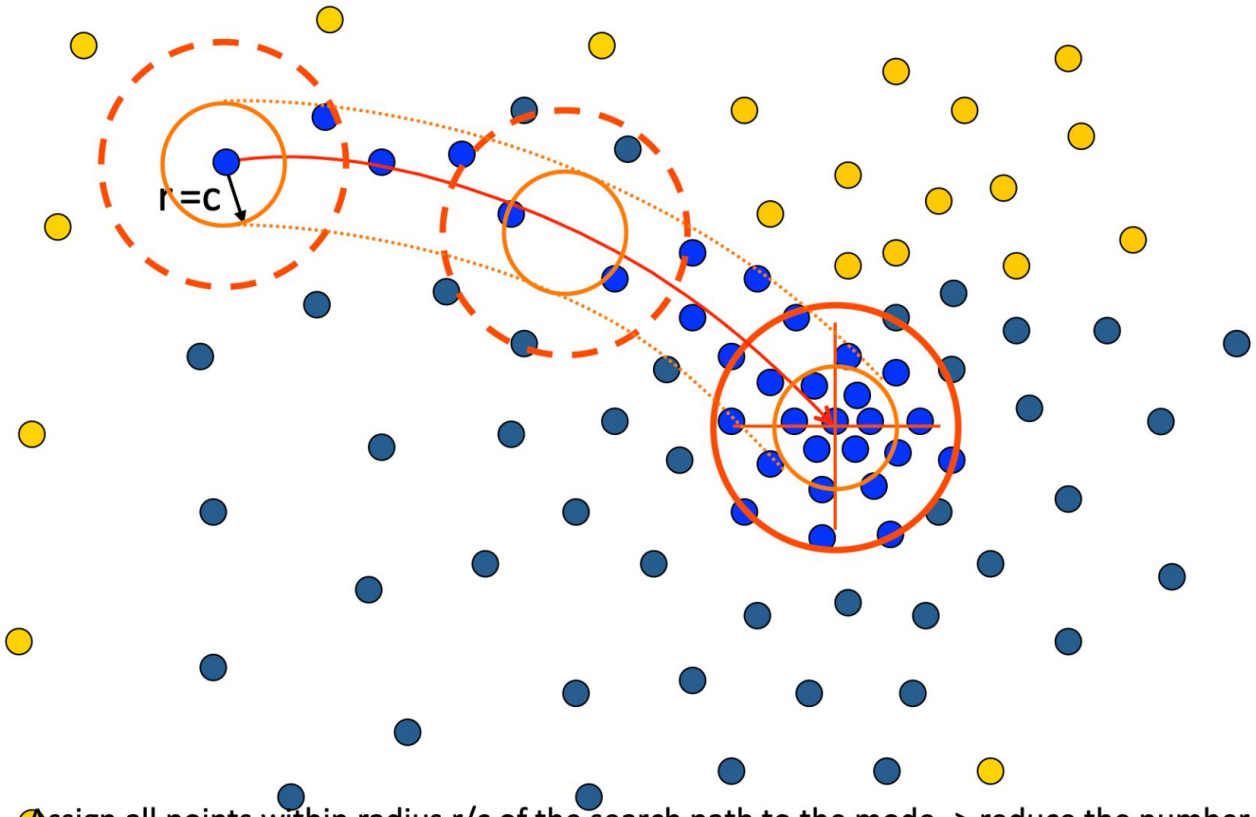
Speedups: Basin of Attraction



1. Assign all points within radius r of end point to the mode.

Slide credit: Bastian Leibe

Speedups



2. Assign all points within radius r/c of the search path to the mode -> reduce the number of data points to search.

Slide credit: Bastian Leibe

Выводы

- Плюсы:
  - Универсальный инструмент, независимый от приложения.
  - Не предполагает форму кластеров (сфера, эллипс и др.).
  - Один параметр - размер окна h (имеет физический смысл).
  - Находит переменное количество мод.
  - Устойчив к выбросам.
- Минусы:
  - Результат зависит от размера окна.
  - Выбор размера окна нетривиален.
  - Вычислительно затратен.
  - Плохо масштабируется с размерностью пространства признаков.

Mean-shift Algorithm Comaniciu & Meer, 2002

Given n data points $\mathbf{x}_i \in \mathbb{R}^d$, the multivariate kernel density estimate using a radially symmetric kernel[1] (e.g., Epanechnikov and Gaussian kernels), $K(\mathbf{x})$, is given by,

$$\hat{f}_K = \frac{1}{nh^d} \sum_{i=1}^{n} K\left(\frac{\mathbf{x}-\mathbf{x}_i}{h}\right), \tag{1}$$

where h (termed the *bandwidth* parameter) defines the radius of kernel. The radially symmetric kernel is defined as,

$$K(\mathbf{x}) = c_k k(\|\mathbf{x}\|^2), \tag{2}$$

where c_k represents a normalization constant. Taking the gradient of the density estimator (1) and some further algebraic manipulation yields,

$$\nabla \hat{f}(\mathbf{x}) = \underbrace{\frac{2c_{k,d}}{nh^{d+2}} \left[\sum_{i=1}^{n} g\left(\left\|\frac{\mathbf{x}-\mathbf{x}_i}{h}\right\|^2\right)\right]}_{\text{term 1}} \underbrace{\left[\frac{\sum_{i=1}^{n} \mathbf{x}_i g\left(\left\|\frac{\mathbf{x}-\mathbf{x}_i}{h}\right\|^2\right)}{\sum_{i=1}^{n} g\left(\left\|\frac{\mathbf{x}-\mathbf{x}_i}{h}\right\|^2\right)} - \mathbf{x}\right]}_{\text{term 2}}, \tag{3}$$

where $g(x) = -k'(x)$ denotes the derivative of the selected kernel profile. The first term is proportional to the density estimate at $\mathbf{x}$ (computed with the kernel $G = c_g g(\|\mathbf{x}\|^2)$). The second term, called the *mean shift* vector, $\mathbf{m}$, points toward the direction of maximum increase in density and is proportional to the density gradient estimate at point $\mathbf{x}$ obtained with kernel K. The mean shift procedure for a given point $\mathbf{x}_i$ is as follows: (see Fig. 1):

1. Compute the mean shift vector $\mathbf{m}(\mathbf{x}_i^t)$.
2. Translate density estimation window: $\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{m}(\mathbf{x}_i^t)$.
3. Iterate steps 1. and 2. until convergence, i.e., $\nabla f(\mathbf{x}_i) = 0$.

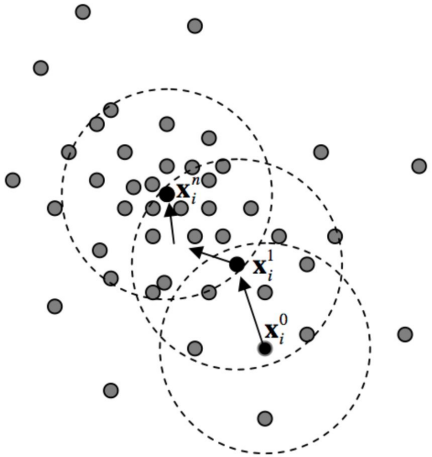


Figure 1: Mean shift procedure. Starting at data point $\mathbf{x}_i$, run the mean shift procedure to find the stationary points of the density function. Superscripts denote the mean shift iteration, the shaded and black dots denote the input data points and successive window centres, respectively, and the dotted circles denote the density estimation windows.